

Jack Xiuqi Zhang

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EDUCATION

Imperial College London

London, UK

MEng Computing, 3rd Year

2023 – 2027

- Achieved High First Class in both first and second year.
- First-year modules: Object Oriented Programming, C Programming, Databases, Calculus, Linear Algebra.
- Second-year modules: Software Development, OS, Compilers, Networks, Algorithms, Probability and Statistics.

Alice Smith School

Kuala Lumpur, Malaysia

A Levels & IGCSEs

2019 – 2023

- A Levels: Mathematics (A*), Further Mathematics (A*), Physics (A*), Computer Science (A)
- IGCSEs: A* + 999998887

EXPERIENCE

Goodnotes

Jun. 2025 – Sep. 2025

Machine Learning and Software Engineer Intern

London, UK

- Owned end-to-end AI-powered outline generation feature from PRD to production, targeting **2M+ users**.
- Self-taught **Swift/iOS development** and built production-ready frontend integrated with backend APIs.
- Implemented a new Python processing pipeline within the existing **FastAPI + AWS Bedrock** backend.
- Developed an experimental **rule-based sentiment analysis system** for quick action suggestions.
- Fixed a critical JSON bug affecting **50k+ daily users**, reducing error rates from **40% to near-zero**.
- Enhanced CI/CD reliability by adding **pytest** retries, lowering pipeline failure rate from **30% to <5%**.
- Designed compliance-ready prompts for the China endpoint, now serving **750k weekly users**.

KEATH.AI

Jun. 2024 – Sep. 2024

Software Engineer and Prompt Engineer Intern

Hybrid

- Engineered a comparative marking system with education specialists, improving AI grading accuracy by **50%**.
- Developed a scalable code evaluation web platform enabling automated grading for programming assignments.
- Built a RAG-based chatbot using Flask + OpenAI API, reducing staff workload by automating Q&A.

PROJECTS

WACC Compiler | Scala, Intel x86

Jan. 2025 – Apr. 2025

- Led 4-person team to develop a **full-featured compiler** for WACC language with external **C FFI** integration.
- Built multi-stage pipeline: lexical analysis, syntax parsing, semantic checking, and Intel x86 code generation.
- Engineered 100+ test suite covering valid programs and error cases for correctness and robustness.
- Integrated GitLab CI pipelines for automated testing and static analysis, enforcing code quality standards.

Pintos Operating System | C

Sep. 2024 – Dec. 2024

- Directed team of 4 in implementing key OS components: **threading, system calls, and virtual memory**.
- Designed and implemented a **priority donation algorithm** to prevent priority inversion in the kernel.
- Developed secure system call handlers for process management with strict kernel/user isolation.
- Implemented synchronization primitives for process execution and thread coordination.
- Architected virtual memory management features: frame allocation and eviction (clock algorithm), lazy loading, page sharing, and memory mapping.

First Year C Project | C, WebSockets

May. 2024 – Jun. 2024

- Designed and implemented an emulator and assembler for the Armv8 AArch64 instruction set.
- Built a WebSocket-based chess server and client with secure handshake protocols and real-time gameplay.

TECHNICAL SKILLS

Programming Languages: Python, C, Java, Scala, Swift, TypeScript, Kotlin, SQL, Haskell

Frameworks & Libraries: FastAPI, PyTorch, TensorFlow, React, SwiftUI/UIKit, NumPy, Pandas, OpenCV

Developer Tools: Git, Linux/Bash, Docker, AWS (Cloud Services), Xcode, GDB

Human Languages: English (native), Mandarin (native)